PCBC H Parametrics Practice Test A Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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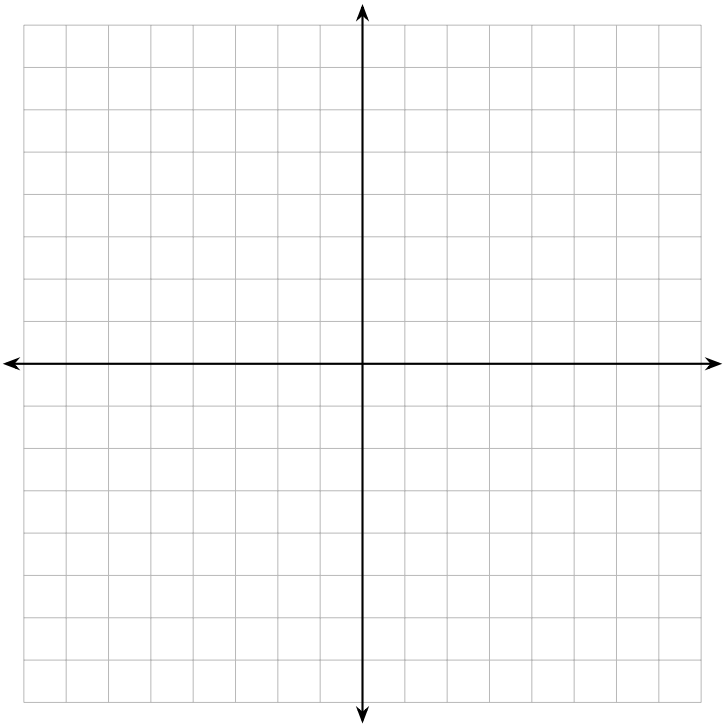
Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

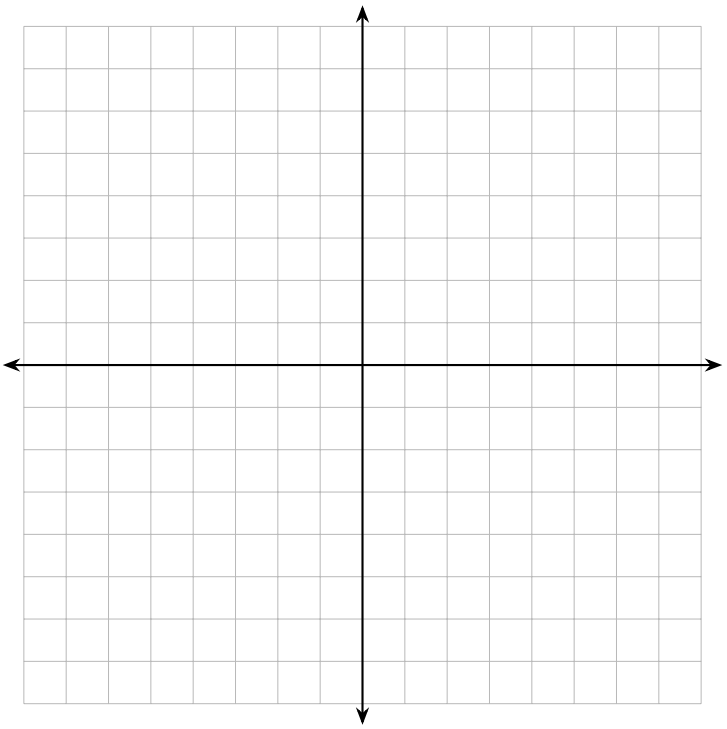
(a) Find the point that corresponds to $t = 0$. (b) Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = \sqrt{t+1}, \quad y = 2t, \quad$ for t in $[-1, 3]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 4\tan t, \quad y = 2\sec t, \quad \text{for } t \text{ in } \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

3. $y = x^2 - 4x + 1$ **3 pts**

Representation 1 *Representation 2*

Calculator | 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

A baseball is hit from a height of 3 ft at an angle of inclination of 35° with a velocity of 110 ft/sec. (a) Write the parametric equations that model its path. (b) For how many seconds is the baseball in flight, and what horizontal distance does it travel? (c) What is its maximum height? **3 pts**

PCBC H Parametrics Practice Test A: Answer Key

12 points

Every value was checked in Python (sympy and 30-digit arithmetic). Work is on the left, the reason for each line is on the right. Final answers are written the way the homework answer keys write them: solved for y when possible, hyperbola branches as $y = \sqrt{\cdot}$, ellipses in "$= 1$" form.

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1. $x = \sqrt{t+1},\ y = 2t,$ for t in $[-1, 3]$ *Non-Calculator*

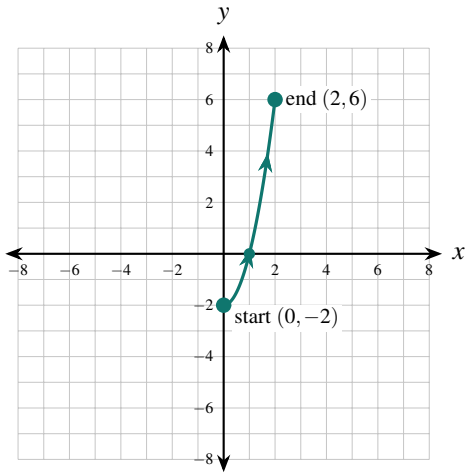
(a) $\boxed{(1,0)}$ (b) $\boxed{y = 2x^2 - 2, \quad 0 \le x \le 2}$

Graph: right half of the parabola from $(0,-2)$ to $(2,6)$, traced up and to the right.

Work	Why
(a) $x = \sqrt{0+1} = 1,\ \ y = 2(0) = 0$	Plug $t = 0$ into both equations.
(b) $x^2 = t + 1$, so $t = x^2 - 1$	Square x, then solve for t. $x = \sqrt{\ } \ge 0$.
$y = 2(x^2 - 1) = 2x^2 - 2$	Substitute into y.
$t = -1 \Rightarrow (0,-2);\ t = 3 \Rightarrow (2,6)$	Endpoints.
$0 \le x \le 2$	$x = \sqrt{t+1}$ only increases, so the t endpoints give the x endpoints.
solid dots at $(0,-2)$ and $(2,6)$	Closed interval, two different end points.

t	(x, y)
-1	$(0,-2)$
0	$(1,0)$
$\frac{5}{4}$	$(\frac{3}{2}, \frac{5}{2})$
3	$(2,6)$

Only the right half: a square root is never negative.



Practice scoring (3 pts): 1 for the point $(1,0)$ and the equation, 1 for $0 \le x \le 2$, 1 for the graph with end dots and direction arrows.

Trap: drawing the whole parabola $y = 2x^2 - 2$. $x = \sqrt{t+1}$ is never negative, and t stops at 3.

2. $x = 4\tan t,\ y = 2\sec t,$ for t in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ *Non-Calculator*

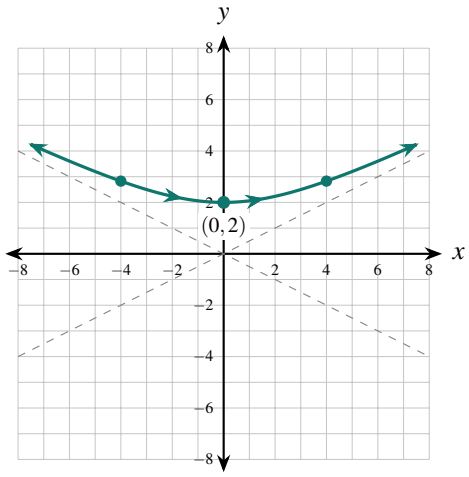
$\boxed{y = \sqrt{\frac{x^2}{4} + 4}}$ (upper branch of $\frac{y^2}{4} - \frac{x^2}{16} = 1$, so $y \geq 2$)

x in $(-\infty, \infty)$. Traced left to right: down to $(0, 2)$ at $t = 0$, then up.

Work	Why
$\tan t = \frac{x}{4}, \quad \sec t = \frac{y}{2}$	Isolate each trig function.
$\sec^2 t - \tan^2 t = 1$	Pythagorean identity ($\tan^2 t + 1 = \sec^2 t$).
$\frac{y^2}{4} - \frac{x^2}{16} = 1$	Substitute and square.
$\sec t > 0$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	Cosine is positive there, so $y > 0$.
$y = +\sqrt{\frac{x^2}{4} + 4}$	Solve for y; positive root only. Lowest point $(0, 2)$.
$y = \pm\frac{1}{2}x$	Asymptotes, slope $\pm\frac{a}{b} = \pm\frac{2}{4}$. Dashed guides only.

t	(x, y)
$-\frac{\pi}{4}$	$(-4, 2\sqrt{2})$
0	$(0, 2)$
$\frac{\pi}{4}$	$(4, 2\sqrt{2})$

$2\sqrt{2} \approx 2.83$. $\tan t$ runs from $-\infty$ to ∞, so the curve goes left to right; arrowheads at the ends mean it keeps going.



mid-curve arrows: direction
end arrowheads: curve continues

Practice scoring (3 pts): 1 for $y = \sqrt{x^2/4 + 4}$ (or the "$= 1$" form with $y \geq 2$), 1 for keeping only the top branch, 1 for the graph with left-to-right arrows and asymptote guides.

Trap: drawing both branches. $\sec t$ is never negative on this interval.

3. $y = x^2 - 4x + 1$ *Non-Calculator*

$\boxed{x = t,\ y = t^2 - 4t + 1,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t + 2,\ y = t^2 - 3,\ \text{for } t \text{ in } (-\infty, \infty)}$

Work	**Why**
Representation 1: $x = t$	Let x be the parameter.
$y = t^2 - 4t + 1,\ \ t$ in $(-\infty, \infty)$	Every real x is allowed, so every real t.
Representation 2: $y = (x^2 - 4x + 4) - 4 + 1$	Complete the square: half of -4 is -2, $(-2)^2 = 4$.
$y = (x - 2)^2 - 3$	Vertex form, vertex $(2, -3)$.
$t = x - 2 \ \Rightarrow\ x = t + 2$	Let t be the inside of the square.
$y = t^2 - 3,\ \ t$ in $(-\infty, \infty)$	Substitute.
Check: $(t + 2)^2 - 4(t + 2) + 1 = t^2 - 3$	Put $x = t + 2$ back into the original.

Practice scoring (3 pts): 1 for each correct pair, 1 for the t interval on both.

Trap: $t = x - 2$ means $x = t + 2$, not $t - 2$.

4. *Calculator*

(a) $\boxed{x = (110\cos 35^\circ)t, \quad y = -16t^2 + (110\sin 35^\circ)t + 3}$
(b) in flight $\boxed{\approx 3.990 \text{ sec}}$, travels $\boxed{\approx 359.555 \text{ ft}}$ (c) maximum height $\boxed{\approx 65.200 \text{ ft}}$

Work	**Why**
(a) $x = (110\cos 35^\circ)t \approx 90.107\,t$	Horizontal: $v_0 \cos\theta$ times t.
$y = -16t^2 + (110\sin 35^\circ)t + 3 \approx -16t^2 + 63.093\,t + 3$	Vertical: gravity, $v_0 \sin\theta$ times t, plus the height.
(b) $0 = -16t^2 + (110\sin 35^\circ)t + 3$	Lands when $y = 0$.
$t = \dfrac{-63.093 \pm \sqrt{3980.778 + 192}}{-32}$	Quadratic formula, $a = -16$, $b = 110\sin 35^\circ$, $c = 3$.
$t \approx 3.990$ or $t \approx -0.047$	Reject the negative time.
$x = (110\cos 35^\circ)(3.99033\ldots) \approx 359.555$ ft	Unrounded t into x.
(c) $t = \dfrac{110\sin 35^\circ}{32} \approx 1.972$	Top of the path: halfway point of the $-16t^2 + bt$ part, $t = \frac{b}{32}$.
$y = -16(1.97167\ldots)^2 + (110\sin 35^\circ)(1.97167\ldots) + 3 \approx 65.200$ ft	Plug the unrounded t into y (or trace the top).

Calculator setup and check

Degree mode. Graph check: $x1(t) = 110\cos(35^\circ)t$, $y1(t) = -16t^2 + 110\sin(35^\circ)t + 3$, $0 \le t \le 4.2$, window x: 0 to 400, y: -5 to 75. Trace the top ($y \approx 65.2$) and where y crosses 0 ($x \approx 359.6$).

Practice scoring (3 pts): 1 for the equations, 1 for flight time and distance, 1 for the maximum height.

Trap: rounding t to 3.99 before finding x gives ≈ 359.526 ft. Keep the full t.